Grothendieck's Dessins d'enfants

A chapter of Master's thesis

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1 Introduction

In this chapter we introduce the main objects of this thesis, called "Dessin d'enfant", after Grothendieck's proposal [5]. There are diferent ways of characterizing these graphs, some more algebraic and some more geometric in nature and this makes these graphs a kind of strong computational tool. We will study some of these corrospondences in this chapter.
There is a great history about the embedding of graphs on surfaces (sometimes called "Topological Graph Theory" for more combinatorial usage) which goes back to Klein and Euler, and Grothendieck's dessins may be considered as a continuation of this history, influenced by Belyi's theorem. Grothendieck's dessins provide a geometric explanation of $\mathbb{G} = Gal(\bar{\mathbb{Q}}/\mathbb{Q})$ by a faithful action which will be explained. Sudying the invariants of this action is an on-going research topic and provides a deep connection between dessin d'enfant and number theory (and the theory of algebraic curves). Some of these examples and computations are explained in this chapter.
We will cover first 4 chapters of [8] (which is the main reference for this thesis) and two great references for this chapter will be [4] and [11] which we will use a lot.

2 Belyi's Theorem and Coverings

In this section we will discuss Belyi's theorem, using concepts from covering spaces(from chapter 1). Let's recall some notions and an important theorem:
Let $\mathbb{P}^1(\mathbb{C})$ be the Riemann sphere. An object of primary interest throughout this chapter will be the Riemann surface obtained by removing the three points $\{0, 1, \infty\}$, that is $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$. Note that for our present purposes, any three distinct points on the sphere would suffice, although later it will be necessary that these three points are actually rational.

Proposition 1. *Let π_1 denote the fundamental group of $\mathbb{P}^1(\mathbb{C})\backslash\{0, 1, \infty\}$. π_1 is generated by loops l_0, l_1 and l_∞ around each of the missing points, with $l_0 l_1 l_\infty = 1$ and so it is nothing but F_2 free groups with two generators.*

The following definition is also central in this chapter:

Definition 1. *Consider the pairs (X, f), where X is a compact Riemann surface and the map $f : X \to \mathbb{P}^1(\mathbb{C})$ is a holomorphic map unramified outside $\{0, 1, \infty\}$. For reasons soon to be explained, such pairs will be referred to as Belyi pairs and the functions f, as Belyi functions. A morphism from one pair (X, f) to another (X', g) is given by a holomorphic map $\rho : X \to X'$ such that $g \circ \rho = f$.*

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, dashed, "\rho"'] & \mathbb{P}^1(\mathbb{C}) \\ X' \arrow[ur, "g"'] & \end{tikzcd}$$

We start discussing the importance of Belyi Pairs with the following Theorem:

Theorem 1. *There is a bijection between conjugacy classes of subgroups of finite index in π_1 and isomorphism classes of Belyi pairs (X, f).*

Proof. Firstly, if we consider $\mathbb{P}^1(\mathbb{C})\backslash\{0,1,\infty\}$ merely as a topological surface, then we have the standard bijection between conjugacy classes of subgroups of π_1 and classes of unramified coverings of $\mathbb{P}^1(\mathbb{C})\backslash\{0,1,\infty\}$, where subgroups of finite index correspond to finite coverings.(This is just the classification of covering spaces, see for example [6]). To complete the connection, note that starting with a Belyi pair (X, f), then considering the restriction of f to $Y = f^{-1}\left(\mathbb{P}^1(\mathbb{C})\backslash\{0,1,\infty\}\right)$ and forgetting about the complex structures, gives the required unramified covering. For the converse, we start with an unramified covering $f : Y \to \mathbb{P}^1(\mathbb{C})\backslash\{0,1,\infty\}$. The complex structure on $\mathbb{P}^1(\mathbb{C})\backslash\{0,1,\infty\}$ induces a unique complex structure on Y. There is a unique way up to isomorphism of Y to compactify Y and to extend f to the compactification X to give a branched covering of $\mathbb{P}^1(\mathbb{C})$ unramified outside $\{0,1,\infty\}$ (This is a standard theorem in Riemann surfaces, see [3]). This gives the corresponding Belyi pair (X, f), completing the proof. □

The main theorem of this section will be the following which shows importance of Belyi Pairs:

Theorem 2. *(Belyi) Let X be a nonsingular projective algebraic curve defined over $\mathbb{C}$. Then X can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a non-constant morphism named $f : X \to \mathbb{P}^1(\mathbb{C})$ unramified outside $\{0,1,\infty\}$, i.e. if and only if there exists an f such that (X, f) is a Belyi pair.*

Proof. By a classical result following from Weil's criterion [20], we know that X can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a morphism $f : X \to \mathbb{P}^1(\mathbb{C})$ with all critical values contained in $\overline{\mathbb{Q}}$. Since $\{0,1,\infty\} \subset \overline{\mathbb{Q}}$, this proves the one direction.
The proof of the other direction is due to Belyi [1] and proceeds along the following lines (following the exposition given by Schneps [17]): Starting with a curve X defined over $\overline{\mathbb{Q}}$, we can find a morphism $f : X \to \mathbb{P}^1(\mathbb{C})$ with all critical values in $\overline{\mathbb{Q}}$ as mentioned above or simply by taking a non-constant element of the function field of X. Now the proof is divided into two steps. In the first we compose f on the left with a polynomial defined over $\mathbb{Q}$ to get a function with all its finite critical values lying in $\mathbb{Q}$ instead of just $\overline{\mathbb{Q}}$. This function is then composed on the left with a rational function defined over $\mathbb{Q}$ such that the resulting function has only two finite critical values, both in $\mathbb{Q}$, which can be taken to be 0 and 1. This function is then still defined on the original X, thus completing the proof.
In more detail: Consider the set $\mathcal{C}$ of all finite critical values of the original function f as well as their conjugates under the action of $\mathbb{G}$ (this action is explained more in next sections, but in fact it is nothing more than acting the permutations on the coefficients). Let g_0 be a polynomial with rational coefficients vanishing exactly on $\mathcal{C}$ (we can take g_0 to be the product of the minimal polynomials of the elements of $\mathcal{C}$, one for each $\mathbb{G}$orbit). Now the finite critical values of $g_0 \circ f$

will in general consist of the image of the critical values of f under g_0 (in this case just 0) as well as the critical values of g_0. Then we repeat the process to find a polynomial g_1 vanishing on these critical values and having degree strictly smaller than g_0 (for example by taking g_1 to be the derivative of g_0). Continuing in this manner we find a g_n with degree 0 for some n. The polynomial $h = g_{n-1} \circ g_{n-2} \circ \cdots \circ g_1 \circ g_0 \circ f$ then has all critical values in $\mathbb{P}^1(\mathbb{Q})$.
Now let $\mathcal{C}'$ be the set of finite critical values of h. If $\mathcal{C}'$ is empty we are done. If $\mathcal{C}' = \{\alpha\}$, then composing h with the linear fractional transformation $z \mapsto z - \alpha$ gives the required function. If $\mathcal{C}' = \{\alpha, \beta\}$, then composing h with the linear fractional transformation $z \mapsto \frac{1}{\beta-\alpha} z + \frac{\alpha}{\alpha-\beta}$ maps the ordered triple (α, β, ∞) to the ordered triple $(0, 1, \infty)$, so again we are done. If $\mathcal{C}' = \{\alpha, \beta, \gamma, \ldots\}$, then the previous transformation will take γ to a rational number of the form $\frac{m}{m+n}$. Composing with the rational function

$$z \mapsto \frac{(m+n)^{(m+n)}}{m^m n^n} z^m (1-z)^n$$

takes 0 and 1 to 0 , and $\frac{m}{m+n}$ to 1 , without introducing new ramification values, thereby giving a function with fewer ramification values than the original h. Repeating this process gives the required function after a finite number of steps. $\square$

3 Definition of Dessins

Intuitively, dessins are bipartite graphs drawn on compact surfaces. In this section we will make this definition more precise, and show how the same objects can be described in many different ways.

3.1 Topological View

In this section we will define a dessin as a purely topological structure. To do so, we start with a 2-dimensional, compact, connected, oriented surface X. Note for future reference that these are precisely the types of surfaces that can be equiped with the analytical structure of a compact Riemann surface.
Now we want to give X a 2-dimensional cell complex structure. This involves partitioning X into three parts, the first (X_V) being a finite discrete set of points (0-cells, or vertices), the second (X_E) homeomorphic to a finite disjoint union of open intervals (1-cells, or edges) and the last is (X_F) homeomorphic to a finite disjoint union of open disks (2-cells, or faces).
Then some consideration will show that the closure of an edge must be homeomorphic to either a circle or a closed interval, and contain either one or two vertices corresponding to these two cases. Recall that a graph is defined by a set of vertices and a set of its subsets, each subset having cardinality one or two. Thus there is a natural way to define a graph G_X associated to X with vertices being the elements of X_V, edges the connected components of X_E and the vertices at the

end of an edge being the ones contained in its closure.
G_X is required to be a connected bipartite graph. Being bipartite means that we can partition the set of vertices X_V into two sets such that every edge lies between a vertex from the one set and a vertex from the other set. In other words, there exists a function $\mu : X_V \to \{0, 1\}$ assuming different values on adjacent vertices. This implies for example that G_X has no loops. We will say a vertex is of type 0 or 1.

Definition 2. *A Dessin $\mathcal{C}$ is defined by the data $(X = X_V \cup X_E \cup X_F, \mu)$ satisfying the above properties. G_X is refered to as the associated graph and X the underlying surface. Let $\mathcal{C}'$ be another dessin with data $(X' = X'_V \cup X'_E \cup X'_F, \mu')$. We say $\mathcal{C}$ is isomorphic to $\mathcal{C}'$ if there exists an orientation preserving homeomorphism $\varphi : X \to X'$ which induces homeomorphisms $X_V \cong X'_V, X_E \cong X'_E$ and $X_F \cong X'_F$, and which respects the bipartite structure by satisfying $\mu = \mu' \circ \varphi$ on X_V.*

We will sometimes adopt the convention of speaking of a dessin as a graph, implying of course the associated graph. It is important to note however that although two isomorphic dessins necessarily have isomorphic associated graphs, the converse is not true since the way the graph is embedded into the surface is also important. In fact, as we will see in the next section, a dessin can be completely described be giving its graph and a cyclic ordering of the edges at each vertex corresponding to their ordering on the underlying surface. Figure 1 illustrates some examples of

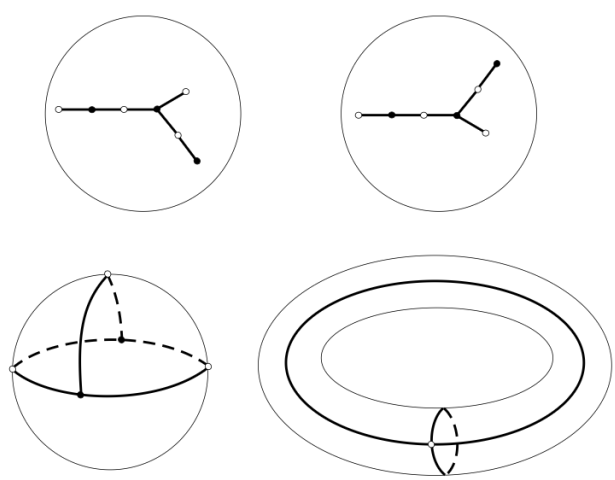

Figure 1: Some examples of dessins

dessins. Notice that the two dessins at the top are isomorphic as graphs but not as dessins.
We will see later that the above description corresponds to taking the pre-image of the unit interval under a function $f : X \to \mathbb{P}^1(\mathbb{C})$. It will sometimes be more convenient however to take the pre-image of the real line, leading to a slightly different description of a dessin as a triangulation of a

surface.

More precisely, consider a dessin $\mathcal{C}$ as described above. Enlarge the set of vertices X_V by adding one point in each face. Then extend the set X_E by adding an edge joining each new vertex to each old vertex bordering on the face in which the new vertex was placed. Also extend the function on the vertices to $\mu : X_V \to \{0, 1, \infty\}$ by letting it take the value ∞ on the new vertices.

On the right, figure 2 shows the new vertices as stars and the new edges as dotted lines. Vertices

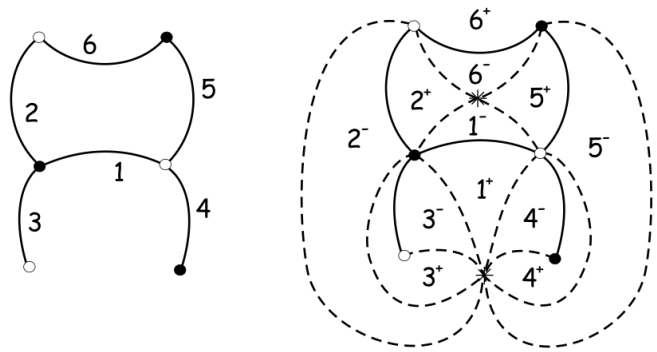

Figure 2: dessins and their triangulation

of type 0 and 1 are indicated by white and black circles respectively.

If n was the number of edges of the original dessin, then there will now be $3n$ edges and $2n$ faces. Each face will be bounded by exactly three edges, hence we have a triangulation of the surface X. The vertices have already been partitioned into three types by means of the values assigned to them by μ. Similarly, we can divide the edges into three types according to the types of vertices linked by the edge, namely $\overline{01}, \overline{0\infty}$ and $\overline{1\infty}$. Finally, the faces, or triangles, can divided into two types according to whether the vertices taken in an anticlockwise direction are arranged $(0, 1, \infty)$, or $(0, \infty, 1)$. We call the triangles positive or negative corresponding to these two cases. Two triangles are said to have the same parity if and only if they are both positive or both negative. Clearly two triangles can only share an edge if they are of different parity.

To go in the reverse direction is now easy: given a triangulation as above, we simply disregard the vertices of type ∞ and the edges of type $\overline{0\infty}$ and $\overline{1\infty}$ to get the original dessin. To summarize, we have shown the following:

Proposition 2. *A dessin can also be described by giving a cell complex $X = X_V \cup X_E \cup X_F$ where every face is bounded by exactly three edges, and a function $\mu : X_E \to \{0, 1, \infty\}$ assuming different values on adjacent vertices. An isomorphism is described exactly as before.*

We conclude this section with a construction which will be useful in the next section. Consider a usual dessin where the n edges have been labelled from 1 to n. Taking the corresponding triangulation, we have a labelling of the edges of type $\overline{01}$. Since each triangle has exactly one

such edge on its boundary, and each such edge is on the boundary of exactly one positive and one negative triangle, we can label the $2n$ triangles as $\{1^+, 1^-, 2^+, 2^-, \ldots, n^+, n^-\}$, where for example triangle $k^+$is the positive triangle having edge k on its boundary. This is illustrated in figure 2.

3.2 Group Theoretic View

All the above information can be codified in a visually less appealing, yet more concise form using groups. Let $F_2 := \langle x, y, z \mid xyz = 1\rangle$ be the free group on two generators x and y. Let F_2 act on the set of edges E of the dessin as follows: x permutes the edges cyclically around each vertex of type 0 in an anticlockwise direction, while y does the same around the vertices of type 1. For example, the permutations induced on the edges of the dessin in figure 2 would be (145)(26)(3) and (123)(4)(56) by x and y respectively.

Since F_2 is acting on a set E with n elements, we get a group homomorphism from F_2 to S_n, the symmetric group on n elements. The image of F_2 under this map is a subgroup of S_n generated by the permutations induced by x and y (denoted by g_0 and g_1 respectively). Note that the action of F_2 is transitive because the graph of the dessin is connected. Then we have the following alternative description of a dessin:

Proposition 3. *There is a bijection between isomorphism classes of dessins and equivalence classes of pairs (g_0, g_1) with $g_0, g_1 \in S_n$ for some n, such that $\langle g_0, g_1\rangle$ is a transitive subgroup of S_n. Two pairs (g_0, g_1) and (g_0', g_1') are considered equivalent if they are simultaneously conjugate in S_n, i.e. if there is some h in S_n such that $g_i' = h^{-1}g_ih$ for $i = 0, 1$.*

Proof. For the one direction it was shown above how to associate a pair (g_0, g_1) with each dessin. In the process it is necessary to label the edges from 1 to n and a different labelling would give another pair which is conjugate to the original one by the permutation h describing the change in labelling.

For the other direction of the proof it is necessary to reconstruct the dessin from a given pair of permutations (g_0, g_1) where $g_0, g_1 \in S_n$. This will be done by joining together $2n$ triangles along their edges to give a triangulation of a surface which describes a dessin as explained in the previous section.

Label the triangles as $1^+, 1^-, 2^+, 2^-, \ldots, n^+, n^-$ and label the three vertices of each as 0,1 and ∞ in such a way that the vertices of the positive (resp. negative) triangles are arranged as described in the previous section. Use $\left(k^+, \overline{ij}\right)$ to refer to the edge of triangle k^+lying between vertices i and j. Now join the triangles by identifying edges as follows (while orientating the edges to preserve vertex types):

$$\left(k^+, \overline{0\infty}\right) \sim \left(g_0(k)^-, \overline{0\infty}\right)$$
$$\left(k^+, \overline{01}\right) \sim \left(k^-, \overline{01}\right)$$
$$\left(g_1(k)^+, \overline{1\infty}\right) \sim \left(k^-, \overline{1\infty}\right)$$

It is clear from this description that each triangle is joined to exactly three other triangles, all with the opposite parity. Thus we can extend the local orientation of each triangle to a global orientation, so we indeed get a compact, oriented surface. Partitioning the surface into vertices, edges and faces, and defining the natural vertex function μ gives a triangulation description of a dessin as in proposition 2.
It only remains to be checked that the permutations of the edges of dessin correspond to the g_0 and g_1 we started with. This can be verified using the three rules for joining edges given above. Indeed, if we start with an edge of type $\overline{01}$ labelled k and try to determine the cyclic ordering of the edges around its 0-vertex, we find that we need to apply the first two rules alternately, giving the sequence of edges $k, g_0(k), g_0\left(g_0(k)\right), \ldots, k$ as desired. The case for g_1 is similar, completing the proof. $\square$

Given a pair (g_0, g_1), it is possible to deduce many properties of the dessin without having to go through the above process to reconstruct it. If we denote the number of cycles of g_i by $|g_i|$, then $|g_0|$ corresponds to the number of vertices of type 0 and the length of each cycle corresponds to the number of edges emanating from that vertex. A similar statement holds for g_1 and if we define g_∞ as $(g_0 g_1)^{-1}$, then $|g_\infty|$ corresponds to the number of faces of the dessin. The length of each cycle of g_∞ is equal to half the number of edges surrounding the corresponding face. Note that some edges can appear twice on the boundary of a face (for example edges 3 and 4 in figure 2). Passing to the corresponding triangulation, $|g_\infty|$ corresponds to the number of vertices of type ∞. All these properties can be verified for figure 2.
Let $v = |g_0| + |g_1| + |g_\infty|$ denote the total number of vertices of the triangulation, $e = 3n$ the number of edges and $f = 2n$ the number of faces. Then we can calculate the genus g using Euler's formula $2 - 2g = v - e + f$. For example, corresponding to figure 2 we have $g_0 = (145)(26)(3), g_1 = (123)(4)(56)$ and $g_\infty = (1364)(25)$, hence $2-2g = (3+3+2)-3(6)+2(6)$, i.e. $g = 0$, which is what it should be since the underlying surface of this specific dessin is a sphere. This group-theoretic description will be useful later on, but for our immediate purposes we need the following:

Proposition 4. *There is a bijection between isomorphism classes of dessins and conjugacy classes of subgroups of finite index in F_2.*

Proof. To associate a conjugacy class of subgroups of F_2 to a given dessin, consider again the action of F_2 on E, the set of n edges of dessin, as described in the beginning of this section. This action is considered to be a right action. Now fix a specifix edge e, and let $H \subset F_2$ be the stabilizer of e, i.e. H is the set of all elements in F_2 which act trivially on e. Since F_2 acts transitively on E, H is a subgroup of index n of F_2. This is because two elements of F_2 lie in the same coset of H if and only if they both act on e by taking it to the same edge e'. So the number of cosets of H is equal to the number of ways an element of F_2 can act on e, that is, to the number of edges

n. Finally, note that choosing a different edge e gives a conjugate subgroup of F_2, again using the fact that F_2 acts transitively.
For the converse we are given a subgroup H of index n in F_2. Let $\mathcal{C}$ be the set of cosets of H. The group F_2 now acts on $\mathcal{C}$ by taking $a \in F_2$ and $Hb \in \mathcal{C}$ to $Hba \in \mathcal{C}$. This action is clearly transitive, so by denoting the images of the generators x and y of F_2 in $\mathcal{C}$ as g_0 and g_1, we can appeal to proposition 3 to reconstruct a dessin. It only remains to verify that starting with a subgroup of F_2 corresponding to a specific dessin, this procedure will yield the same desssin again, essentially because F_2 acts on the set of cosets in exactly the same way as on the set of edges. More precisely, suppose the H that we started with is the stabilizer of an edge e of some specific dessin. Then we can establish a bijection between the set of edges E of the dessin and the cosets $\mathcal{C}$ by associating an edge e' with the coset Ha where $a \in F_2$ takes e to e' and conversely associating a coset Ha with the image of e in E under the action of a. This bijection commutes with the action of F_2 on E and $\mathcal{C}$ respectively, thus completing the proof. $\square$

4 Grothendieck's Correspondence

The results of the previous sections combine to prove the following theorem, known as the "Grothendieck correspondence":

Theorem 3. *There is a bijection between the isomorphism classes of dessins and the isomorphism classes of Belyi pairs.*

Although we have been working with sets of objects and the concept of an isomorphism between two objects, it would have been possible to define general morphisms between objects and treat the sets of objects as categories. The above correspondence would then have been an equivalence of categories since the bijection respects the morphisms between objects.
As we have seen, the proof of this correspondence takes a detour through group theory. There is however a more explicit way of realizing the correspondence. In the one direction, starting with a Belyi pair (X, f), we can find the corresponding dessin on X simply by taking the preimage of the unit interval under f, i.e. $f^{-1}([0,1])$. The points above 0 will be those of type 0 , and the ones above 1 will be those of type 1 . This construction makes it clear that the number of elements in the fiber above 0 is precisely the number of vertices of type 0 , and order of ramification at each point in the fiber is precisely the number of edges meeting at the corresponding vertex. Furthermore, the action of F_2 on the edges of the dessin can be seen to correspond directly to the action of the fundamental group π_1 on points of X. More precisely, choose $\frac{1}{2}$ as the base point for the fundamental group and consider the fiber above this point. From the above description, it is clear that there is exactly one point in the fiber on every edge of the dessin. Now the fundamental group acts on this fiber via the lifting of loops. The action of the loop l_0 corresponds exactly to

the action of $x \in F_2$ on the edges.

To realize the converse direction more concretely, one must make use of the triangulation associated to a dessin with underlying surface X. Consider the simple triangulation of the sphere by adding the three edges joining the points 0,1 and ∞. There are two triangles, called positive and negative by the original convention. Now, since every triangle is homeomorphic to an open disk, one can define a function f from the triangles of X to the triangles of the sphere, such that f restricted to a single triangle is a homeomorphism between that triangle and the one on the sphere with the same parity. By choosing f appropriately, it is possible to extend it to a continuous function defined on the whole of X, respecting the vertex, edge and face types. This is then the required Belyi function. Using this function we can lift the complex structure on $\mathbb{P}^1(\mathbb{C})$ to a complex structure on X, giving the required Belyi pair.

Note that in the construction just described, one starts with an arbitrary topological surface X, and by drawing a dessin on it, one can equip X with a complex structure. Two questions arise, namely is this complex structure unique, and which complex structures arise in this way? The positive answer to the first question was first pointed out by Grothendieck in [5], but proven essentially before by Jones and Singerman [9] and independently by Malgoire and Voisin [19] (see also the survey by Wolfart [21]). The answer to the second question is Belyi's theorem, given in the previous section.

Also note that although the Riemann surface X associated to a given dessin is unique, the Belyi function f is only unique up to composing with automorphisms of X. Thus in genus 0 the map f is unique up to $\mathrm{PSL}_2(\mathbb{C})$, in genus 1 up to affine transformation and in genus greater than 1, up to a finite automorphism group. To obtain uniqueness, one needs to restrict f in some way. In genus 0 , this can be done by fixing the position of three seperate points on the dessin since the group $\mathrm{PSL}_2(\mathbb{C})$ acts three-transitively on the sphere.

5 Galois Action on Dessins

5.1 The Action and Faithfullness

An element of $\mathbb{G} = \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ can be viewed as a consistent choice of automorphisms of every finite Galois extension. To make $\mathbb{G}$ into a topological group, start by giving the finite Galois groups the discrete topology. The product of these groups is then given the product topology and $\mathbb{G}$ receives the restriction topology as its subgroup. Equivalently, this is the coarsest topology making all the projection maps from $\mathbb{G}$ to the finite Galois groups, continuous. Consequently, a subgroup of $\mathbb{G}$ is open if and only if it has finite index in $\mathbb{G}$.

Since we can view dessins as Belyi pairs (X, f) with both X and f defined over $\overline{\mathbb{Q}}$, it is natural to consider the action of $\mathbb{G}$ on the dessins. An element $\sigma \in \mathbb{G}$ acts on a pair (X, f) by acting on

the coefficients of a defining equation of X and on the coefficients of f. We denote the resulting pair by (X^σ, f^σ).
The question then arises as to whether the action is faithful, that is, whether for any given $\sigma \in \mathbb{G}, \sigma \neq 1$, one can always find a dessin on which σ acts non-trivially. That this is indeed the case follows from the following:

Proposition 5. *The action of $\mathbb{G}$ on the set of genus 1 dessins is faithful.*

Proof. Let σ be an element of $\mathbb{G}$. Choose any algebraic number j on which σ acts non-trivially. Let X be an elliptic curve with j-invariant equal to j. Consider any nonconstant function on X, and using the process described in the proof of Belyi's theorem, compose it with a series of functions to produce a Belyi function f on X. Since X^σ has j-invariant equal to $\sigma(j), \sigma$ acts non-trivially on the pair (X, f). □

The action on the set of genus 0 dessins is also faithful. Indeed, consider the set of genus 0 dessins having only one open cell, that is, only one point above ∞. These dessins are called trees, because by choosing this one point to be ∞, the dessin can be realised as a planar graph with no cycles. Then a result by Lenstra (proved in [18]) states that the action of $\mathbb{G}$ on the set of trees is faithful.
Thus each element of $\mathbb{G}$ can be identified with a unique permutation of the set of dessins, so a better understanding of this action could provide new insight into $\mathbb{G}$. A first step would be to characterize the orbit of a dessin under the group action. By a result of the next section, such an orbit is always finite. However it remains an open problem to decide whether two given dessins lie in the same Galois orbit or not, and some approaches will be described in the next part.
Another consequence of Belyi's theorem is that we can attach a unique number field to each dessin, namely the moduli field of the dessin. This is defined to be the field K of the following lemma (adapted from Wolfart [21]):

Lemma 1. *Let (X, f) be a Belyi pair. Then for a field $K \subset \overline{\mathbb{Q}}$ the following properties are equivalent:*
1. K is the minimal field with the property $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K)$ implies $(X, f) \cong (X^\sigma, f^\sigma)$.
2. For all $\sigma \in \mathbb{G}, \sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/K) \Longleftrightarrow (X, f) \cong (X^\sigma, f^\sigma)$.
3. K is the fixed field of $\{\sigma \in \mathbb{G} \mid (X, f) \cong (X^\sigma, f^\sigma)\}$.

We call L **a field of definition of** (X, f) if there is some isomorphic Belyi pair (X', f') with both X' and f' defined over L. Clearly any field of definition contains the moduli field, and in fact the moduli field is the intersection of all possible fields of definition.
One can always find a field of definition which is a number field, hence the moduli field is a number field. From the definition of the moduli field, it follows that the number of elements in the Galois orbit of a given dessin is equal to the degree of its moduli field over $\mathbb{Q}$.

5.2 Galois Invariants

As noted in the previous section, $\mathbb{G} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully on the set of dessins, dividing it into orbits. Ideally, we would like to have a simple (possibly combinatorial) criterion for telling whether two given dessins lie in the same orbit or not. Currently, the only sure way of doing this is computing a Belyi function for a given dessin and using it to find conjugate dessins. There are however several known necessary (but not sufficient) invariants characterizing the Galois orbit of a dessin. This section introduces the simpler ones.
Just as the Galois action partitions the set of dessins into orbits, the Galois invariants also induce partitions which are by definition coarser. We call an invariant finer in general than another one if it induces a finer partition. Sometimes we will refer to an invariant as being finer in some cases than another simply if it isn't coarser in general. Most invariants seem to be finer than all others in some cases, implying that the best method is a combination of them all.

5.2.1 Elementary Invariants

We make the obvious observation that the bijection $X \leftrightarrow X^\sigma$ performed by a Galois element $\sigma \in \mathbb{G}$ is never holomorphic (not even continuous with the exception of the case in which σ is the complex conjugation). Therefore, X and X^σ will in general be nonisomorphic. Likewise there is not an a priori reason why these two Riemann surfaces should be even homeomorphic. Nevertheless, the action $(X, f)^\sigma = (X^\sigma, f^\sigma)$ preserves the genus as well as some other interesting properties:

Theorem 4. *The action of $\mathbb{G}$ on pairs (X, f) enjoys the following properties:*
(1) $\deg(f^\sigma) = \deg(f)$.
(2) $(f(P))^\sigma = f^\sigma(P^\sigma)$.
(3) $\mathrm{ord}_{P^\sigma}(f^\sigma) = \mathrm{ord}_P(f)$.
(4) The genus of X^σ is the same as the genus of X, i.e. X and X^σ are homeomorphic.

Proof. See [4] □

It is also good to think about "What are not invariants" for this action. The simplest answer is the graph itself (and some combinatorial characters of the graph). The examples we illustrate in the section 6 are also useful for this matter.

5.2.2 Valency Type

Consider a dessin $\mathcal{S}$ with n edges.

Definition 3. *The valency of a vertex is the number of edges meeting at the vertex. The valency of a face is half the number of edges bordering on the face, where an edge surrounded by the face is counted twice.*

Since we are dealing with bipartite graphs, the number of edges bordering on a face will always be even, so this definition makes sense. For now we will refer to the faces as vertices of type ∞. Then it is clear from the correspondence between valencies of vertices of type i and orbit lengths of g_i where $i \in \{0, 1, \infty\}$ that the sum of the valencies of vertices of type i equals the number of edges, n.

To make this precise, let m_i be the number of vertices of type $i, i \in \{0, 1, \infty\}$, and label them $1, 2, \ldots, m_i$. Let $k_r^{(i)}$ be the valency of the r 'th vertex of type i. Then for each i there is an unordered partition $n = k_1^{(i)} + k_2^{(i)} + \cdots + k_{m_i}^{(i)}$. We say the dessin has valency type

$$\left(k_1^{(0)} + \cdots + k_{m_0}^{(0)}; k_1^{(1)} + \cdots + k_{m_1}^{(1)}; k_1^{(\infty)} + \cdots + k_{m_\infty}^{(\infty)}\right)$$

For example the valency type corresponding to Figure 2 is $(1 + 2 + 3, 1 + 2 + 3, 2 + 4)$.

It can be shown that the valency type remains invariant under the Galois action (see [13]). Furthermore, there are only a finite number of graphs with a given valency type, hence Galois orbits must be finite. There is no upper bound on their lengths however:

Proposition 6. *Galois orbits of dessins can be arbitrarily large.*

Proof. Let σ be an element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which is not a torsion element. Then for every dessin s there is a positive number n_s such that $\sigma^{n_s}(s) = s$ (since orbits are finite). Suppose orbits are bounded in length by M, i.e. $n_s \leqslant M$ for all dessins s. Now let $n = M!$. Then $\sigma^n(s) = s$ for all s. But since $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts faithfully, it follows that σ^n must be the identity, i.e. σ is a torsion element. The contradiction shows that orbits are not bounded in length. $\square$

The genus of the underlying surface of the dessin is determined by the valency type. Thus from the invariance of the valency type we can also deduce the invariance of the genus.

5.2.3 Monodromy Groups and Composition

Consider the dessin given by the pair of permutations (g_0, g_1) from S_n. The subgroup of S_n generated by g_0 and g_1 is called the monodromy group of the dessin. This group is another Galois invariant, in general finer than the valency type.

Variations of the monodromy group invariant can be produced by first composing the given dessin with another one and then calculating the monodromy group of the resulting dessin. Given Belyi pairs (X, f) and (Y, g), we would like to define their composition as the Belyi pair $(X, g \circ f)$. Clearly certain restrictions have to be placed on the pair (Y, g) for this to make sense, leading to the definition:

Definition 4. *A Belyi pair (Y, g) is said to be composable if $Y = \mathbb{P}^1(\mathbb{C}), g(\{0, 1, \infty\}) \subset \{0, 1, \infty\}$ and g is defined over $\mathbb{Q}$. Given such a pair and another general pair (X, f), this is a sufficient condition for $(X, g \circ f)$ to be a Belyi pair, called the composition of (X, f) with g.*

Since Y is always $\mathbb{P}^1(\mathbb{C})$, we just refer to g as a composable Belyi function. The associated dessin is called a composable dessin. The reason for requiring g to be defined over $\mathbb{Q}$ is to ensure that composition commutes with the Galois action:

Proposition 7. *Let g be a composable Belyi function. Then if two dessins are in the same Galois orbit, their respective compositions with g will also lie in the same Galois orbit.*

Proof. Let (X, f) be a Belyi pair, and σ an element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Then the proof of the theorem will follow from the commutativity of the following diagram, where the horizontal arrows indicate composition with g, and the vertical arrows indicate the action of σ.

$$\begin{tikzcd}
(X, f) \arrow[r, "g"] \arrow[d, "\sigma"] & (X, g \circ f) \arrow[d, "\sigma"] \\
(X, f^\sigma) \arrow[r, "g"'] & (X^\sigma, (g \circ f)^\sigma)
\end{tikzcd}$$

Going around clockwise we find $h = (g \circ f)^\sigma$, while going anti-clockwise gives $h = g \circ f^\sigma$. Since g is defined over $\mathbb{Q}$, these functions coincide. $\square$

An examples of composing dessins is shown in figure 3. The dessin in the middle is composed with the dessin drawn next to the arrows, and the resulting composition is labelled A.

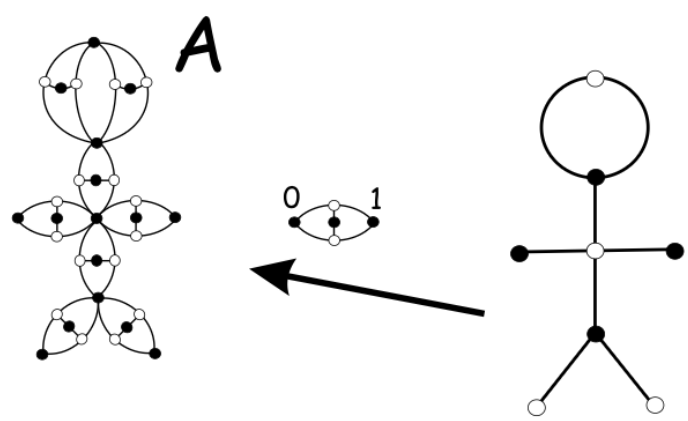

Figure 3: composition of dessins

5.2.4 Trees

Trees are genus 0 dessins with a single face. They are simpler than general dessins in certain respects, such as always having a polynomial as a Belyi function. Yet as noted before, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$

still acts faithfully on the set of trees. Consequently trees are a natural starting point for studying properties of dessins. In this section we state (a corollary of) a result of [22] which gives an invariant which is strictly finer than valency types for a certain family of trees.

For convenience we change the notation from that given at the beginning of the section. Let T be a specific tree. Let n and m denote the number of vertices of type 1 and 0 respectively. Label the vertices of each type, and let p_i and q_j denote the valencies of the i 'th vertex of type 1 and the j 'th vertex of type 0 respectively. We denote the valency type of the tree by $(p_1, \ldots, p_n; q_1, \ldots, q_m)$. From Euler's formula then follows that $\sum_i p_i = \sum_j q_j = m + n - 1$.

Actually it is necessary to label the vertices of type 1 (the p_i 's) in a specific way to capture the way the graph is embedded into the sphere. First note that this embedding is described by giving a cyclic permutation of the edges meeting at each vertex corresponding to their order in an anticlockwise direction around the vertex. Now choose a vertex of type 1 to be labelled as number 1 and choose an edge leading away from it as the active edge. We now traverse the graph using the following algorithm:

Walk along the active edge to the next vertex. Replace the current active edge by the edge following it in the permutation corresponding to this vertex. Now walk along this edge, repeating the process until reaching the original vertex and same active edge. Along the way, label the vertices of type 1 in the order in which they are first reached. Note that every edge will be traversed exactly twice.

Another way to describe it, is to consider the pair of permutations (g_0, g_1) corresponding to the tree. The order in which the edges are traversed can be found by applying g_0 and g_1 alternately.

Having labelled the type 1 vertices, we now make the assumption that they all have distinct valencies, i.e. that the p_i 's are all distinct. There is then a unique permutation σ on n elements such that $p_{\sigma(1)} < p_{\sigma(2)} < \cdots < p_{\sigma(n)}$. Let $S(T)$ (the signature) be either 1 or -1 corresponding to whether this is an even or odd permutation.

Note that $S(T)$ depends on the choice of starting vertex, in the sense that starting at a different vertex will give a different permutation which could have a different signature. However, in the case where n as well as all the q_i 's are odd (conditions which are met in the following theorem), it can be verified that the signature is independant of the choice of starting vertex.

Theorem 5. *Let $n, m, p_1, \ldots, p_n, q_1, \ldots, q_m$ be positive integers such that:*

1. $p_1 < \cdots < p_n$ and $\sum_i p_i = \sum_j q_j = m + n - 1$.

2. All the q_j 's are odd.

3. $n \equiv 1 (\mathrm{mod} 4)$

4. $p_1 \ldots p_n (p_1 + \cdots + p_n)$ is a perfect square.

Then $S(T)$ is a Galois invariant for the trees T having valency type ($p_1, \ldots, p_n; q_1, \ldots, q_m$). Thus this valency class splits into at least two Galois orbits.

One of the examples which started conjectures in this direction is known as Leila's flowers (after

the works of Leila Schneps). They are trees forming a valency class of order 24 which splits into two Galois orbits of equal size. A tree from each orbit is shown in figure ????????????. The vertices of type 1 have distinct valencies $(2, 3, 4, 5, 6)$ and there are $5(\equiv 1(\bmod 4))$ of them. All the vertices of type 0 have valency 1 or 5 , hence odd. Finally $p_1 \dots p_n\,(p_1 + \cdots + p_n) = 14400 = (120)^2$ is a perfect square. Thus the theorem applies. Labelling the vertices of type 1 as indicated shows that the tree on the right needs the odd permutation $(12)(3)(4)(5)$ to sort the valencies in increasing order, whereas for the tree on the left they are already sorted. Since the identity permutation is even, this shows that they are in different orbits.

6 Explicit Examples

In this section, some examples of Belyi maps and calculations on dessins are discussed. The main sources for this section are [11] and [4] and [10]. A huge set of examples of Belyi maps are available at the LMFDB which we explain in the final subsection.

6.1 A Genus 0 Example

Example 1. *As we mentioned earlier, trees are among the simplest (but very rich!) cases to consider as dessins. Trees with 2, 3 and 4 vertices (which are unique) come as dessins by the Belyi map $f : \mathbb{P}^1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C})$, $f(z) = z^n$ for $n = 2, 3, 4$. This map for any n gives a "star shape" dessin which is called "n-star" graph. As the equation is defined over $\mathbb{Q}$, we get a length 1 Galois orbit and the Monodromy group is generated by id and $(1, ..., n)$.*
The smallest non trivial cases in trees to consider, are two other trees with 4 edges: Considering

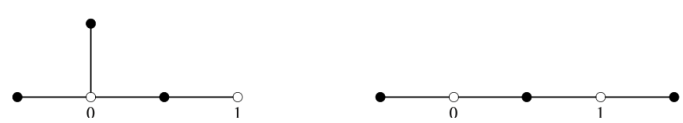

Figure 4: Non trivial trees with 4 edges

the left hand side dessin, a simple calculation shows that the degree 4 polynomial corresponding to $\mathcal{D}$ is:

$$f_{\mathcal{D}}(z) = -\frac{256}{27} z^3 (z - 1)$$

which is the Belyi function associated to this dessin. The permutation representation pair is given by $\sigma_0 = (1, 2, 3), \sigma_1 = (3, 4)$.
Note that both dessins in figure —— are particular cases of the configuration depicted in figure ——— , which we shall call the (m, n)-double star dessin, and whose corresponding Belyi function is of the form $f(z) = C_{m,n} z^m (z - 1)^n$, where the constant $C_{m,n}$ can be easily determined.

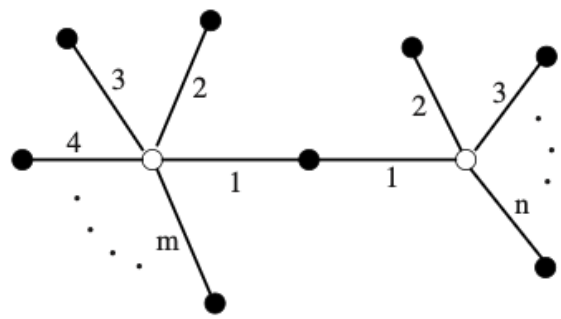

Figure 5: Double star dessins

The following proposition is not hard to prove ([4]):

Proposition 8. *The action of $\mathbb{G}$ on trees with less than five edges is trivial.*

6.2 A Genus 1 Example

Example 2. *Consider the elliptic curve X given by the equation*

$$X : y^2 = x(x-1)(x-\alpha),$$

where we take

$$\alpha := \frac{1}{\sqrt[3]{2}} \in \mathbb{R}$$

Clearly, this curve is defined over $\overline{\mathbb{Q}}$. We need a meromorphic function on X to go through Belyi's theorem, so we start with the projection

$$X \to \hat{\mathbb{C}}, (x,y) \mapsto x$$

which is ramified at the critical points $(\infty,\infty), (0,0), (1,0)$ and $(\alpha,0)$, with critical values $\infty, 0, 1$ and α. In order to find a Belyĭ function, we need to eliminate the critical value α, so we compose this projection with the function $x \mapsto x^3$ which sends the algebraic number α to the rational number $\frac{1}{2}$, while fixing the other three critical values. To deal with the resulting critical value $\frac{1}{2}$, we now apply the function $\beta_{1,1} : z \mapsto 4z(1-z)$ which sends $\frac{1}{2}$ to 1, and sends 0,1 and ∞ to 0,0 and ∞. The composition

$$\beta : (x,y) \mapsto x \mapsto x^3 \mapsto 4x^3\left(1-x^3\right).$$

of these three steps therefore maps the critical values of the first step into $\{0,1,\infty\}$ in the following way:

$$\infty \mapsto \infty, \quad 0 \mapsto 0, \quad 1 \mapsto 0, \quad \alpha \mapsto 1.$$

Using polynomials like this to deal with algebraic critical values may induce new ramifications. Here $x \mapsto x^3$ is ramified at $x = 0$ and $x = \infty$, so the composition $(x, y) \mapsto x^3$ is ramified over $0, 1, \infty$ and $\frac{1}{2}$. However, the last step is ramified only at ∞ and $\frac{1}{2}$, and sends these points to ∞ and 1 , so the composition β is a Belyĭ function $X \to \hat{\mathbb{C}}$. Considering inverse images of ∞, 0 and 1 with their multiplicities, we get the following dessin: Now, let's consider a non-trivia Galois

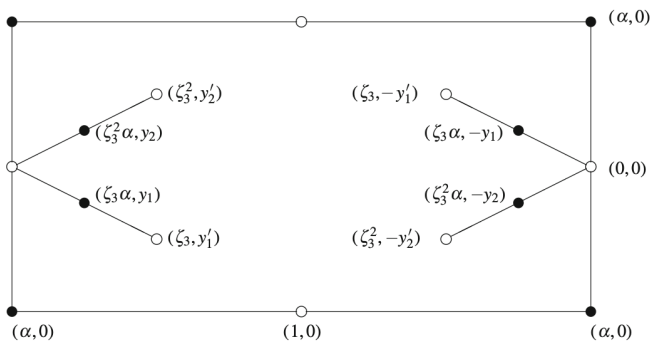

Figure 6: Dessin for Example 2

action $\sigma \in \mathbb{G}$:

$$\sigma(\sqrt[3]{2}) = \zeta_3^{-1}\sqrt[3]{2}$$

Then X^σ is given by

$$y^2 = x(x-1)\left(x - \frac{\zeta_3}{\sqrt[3]{2}}\right)$$

the Belyĭ function is $\beta(x, y) = 4x^3(1 - x^3)$, but the dessin now has the form given in next figure: Note that one has to identify opposite sides of the rectangle to get a dessin on a torus. The Belyĭ

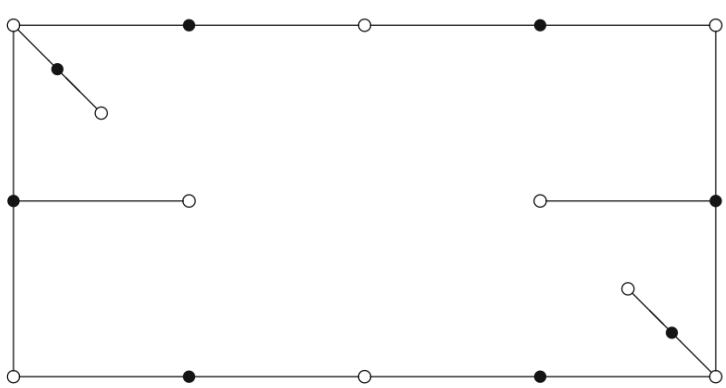

Figure 7: Galois conjugate of figure 6

function is given by the same formula as for the original dessin because it is a polynomial in $\mathbb{Q}[x]$.

6.3 Two Examples From LMFDB

The huge database of LMFDB[1] has been recently used to uncover and understand new patterns in number theoretic objects([7] and [16]). LMFDB also contains 1,111 examples of Belyi maps and highlights some interesting facts and examples which we breifly mention here.

Example 3. *LMFDB highlights this Belyi map as the lowest degree map (degree 4) with hyperbolic geometry type. The given Belyi map is*

$$f(x,t) = \frac{-4}{27}t$$

which is defined on the genus 1 curve (elliptic curve) $t^2 + (2x^2 - 4x + 9)t + x^4 = 0$. *The curve is isomorphic with the minimal Weierstrass equation* $y^2 = x^3 + x^2 + 16x + 180$ *and there is also a page for this curve on LMFDB. This belyi map is defined on* $\mathbb{Q}$ *and the orbit size for the action of* $\mathbb{G}$ *is 1. The monodromy group for this map is* S_4.

Example 4. *The longest Galois orbit on LMFDB belongs to a degree 7 Balyi map (with a huge formula!) which is defined on the curve (with a huge polynomial!) with genus 1. The Galois orbit contains 21 elements (and so the field of definition is a extension of degree 21 !)*

7 Two Applications in Number Theory

In this final section, we mention to applications of Belyi maps and dessins in number theory. The first one is from [11] and the second one is from [10].

7.1 ABC Conjecture

There exist several versions of the ABC problem; here we choose one of them which can be more relavant. Let a, b, c be three positive coprime integers such that $a + b = c$. Note that $c = \max(a, b, c)$. Let

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}, \quad b = q_1^{\beta_1} q_2^{\beta_2} \dots q_m^{\beta_m}, \quad c = r_1^{\gamma_1} r_2^{\gamma_2} \dots r_n^{\gamma_n}$$

be their factorizations into prime factors. Denote by R the product of all the prime factors of a, b, c taken only once, that is,

$$R = p_1 p_2 \dots p_k q_1 q_2 \dots q_m r_1 r_2 \dots r_n.$$

[1] L-functions and Modular Forms DataBase

The conjecture affirms that R cannot be too small compared to c. More precisely: let $c = R^\alpha$, or, equivalently, $\alpha = \log c / \log R$.

Conjecture 1. *(abc). For any $\alpha_0 > 1$ there exist at most a finite number of triples (a, b, c) such that the corresponding value of the parameter α satisfies the inequality $\alpha > \alpha_0$.*

Note that if we replace $\alpha_0 > 1$ with 1 , the statement turns wrong:

Example 5. *Take the equality $1 + (2^{6n} - 1) = 2^{6n}$. The contribution of $a = 1$ and of $c = 2^{6n}$ to R is only 2, while $b = 2^{6n} - 1$ is divisible by $2^6 - 1 = 63$ and hence divisible by 9. Therefore the contribution of b is not bigger than $b/3$, and we conclude that $R < c$, so $\alpha > 1$.*

The next example shows what kind of consequences may be obtained from the conjecture.

Example 6. *Consider the equality $x^n + y^n = z^n$. Obviously,*

$$R \leq xyz < z^3;$$

therefore

$$\alpha = \frac{\log c}{\log R} > \frac{\log z^n}{\log z^3} = \frac{n}{3}.$$

Thus, the abc conjecture implies that all Fermat equations with $n \geq 4$ taken together may have at most a finite number of coprime solutions.

Keeping in mind the conjecture, it is interesting to look for triples a, b, c with the parameter α as big as possible. The world record up to now belongs to Eric Reyssat: $2 + 3^{10} \cdot 109 = 23^5$, with $\alpha = 1.62998$ (rounded).This triple was found by a brute force search. A number of papers were dedicated to a more systematic search: see, for example [14], [15], [2] (which lists a complete list of known instances with $\alpha > 1.4$), [14] (which lists a complete list of instances with the prime factors not greater than 13), etc. A rather long list of instances with $\alpha > 1.2$ was collected **by Noam Elkies** several years ago.
It is clear that in a "good" example the numbers a, b, c must be well factorizable in prime factors. Several authors proposed to use, in a search for good triples, Belyi functions defined over $\mathbb{Q}$. The polynomials constituting a Belyi function are already well factorized. Substituting rational numbers into them, we may, with a little bit of luck, find interesting examples. Let us illustrate it by taking one particular (and very simple) Belyi function:

Example 7. *We take the dessin of example 2.3.1 in [11] though for technical reasons we have changed its position: we put the vertex of degree 1 at infinity, the vertex of degree 3, at 0, and the center of the face of degree 1, at 1. The corresponding Belyi function is*

$$f(x) = \frac{64x^3}{(x+9)^3(x+1)}, \qquad \textit{and} \qquad f(x) - 1 = -\frac{(x^2 - 18x - 27)^2}{(x+9)^3(x+1)}.$$

The above two equalities, with $x = a/b$, may be rewritten as

$$64a^3b + \left(a^2 - 18ab - 27b^2\right)^2 = (a+9b)^3(a+b)$$

Substituting $(a, b) = (-32, 23)$ we get, after rearranging the terms,

$$11^2 + 3^2 \cdot 5^6 \cdot 7^3 = 2^{21} \cdot 23$$

which gives the value of α equal to 1.62599*. This is not the record, but this is the second best value known up to now. However, this triple was found before the search undertaken by N. Magot [12].*

In general, the above Belyi function has produced 43 triples with $\alpha > 1.2$; among them 11 triples were "new" (with respect to the list of Elkies as it was in 1996). The best among the "new" triples was

$$2^{15} \cdot 5 \cdot 67 + 11^2 \cdot 23^8 = 7^3 \cdot 3023^3, \quad \alpha = 1.35670\ldots$$

Several hundred "new" triples with $\alpha > 1.2$ and not present in the Elkies list were found by L. Habsieger.

7.2 Elliptic Curves and Complex Multiplication

We introduce Elliptic curves as tori $T := \mathbb{C}/\Lambda$. Multiplying by a non-zero complex number, we may assume that Λ is the lattice $\mathbb{Z} + \mathbb{Z}\tau$ with period quotient $\tau \in \mathbb{H}$. For any positive integer n, the holomorphic function $\mathbb{C} \to \mathbb{C}, z \mapsto nz$ induces an unramified covering $\alpha_n : T \to T$ which has degree $|\Lambda : n\Lambda| = n^2$. If there is a Belyi function $\beta : T \to \mathbb{P}^1(\mathbb{C})$ of degree d, then it's easy to show that $\beta \circ \alpha_n$ is also a Belyi function, now of degree $n^2 d$. We denote their respective dessins by $\mathcal{D}$ and $n^{-1}\mathcal{D}$, and call $n^{-1}\mathcal{D}$ an **unramified self-cover of $\mathcal{D}$ of degree n^2**. The following easy proposition shows that it is in fact reasonable to speak here about **coverings of dessins**:

Proposition 9. *Show that the dessin $n^{-1}\mathcal{D}$ has a group A of automorphisms isomorphic to $C_n \times C_n$, acting freely on vertices and faces, such that*

$$\mathcal{D} \cong A \backslash n^{-1}\mathcal{D}.$$

Similar to the coverings in topological context, it can be shown that self-covers are always unramified, and that the underlying base surface is isomorphic to the covering surface. One should note that such coverings exist only in genus 1:

Proposition 10. *For genus $g \neq 1$ there are no Riemann surfaces X with unramified coverings $X \to X$ of degree greater than 1.*

The essential point of our consideration is contained in the following:

Theorem 6. *For any dessin on a Belyi pair of genus 1 and for any positive integer n there is a self-cover of degree n^2.*

The algebraic background behind this theorem is the fact that elliptic curves always have $\mathbb{Z}$ as a ring of endomorphisms, where each $n \in \mathbb{Z}$ acts as the endomorphism induced by $z \mapsto nz$; here we again write the elliptic curve as a torus $T = \mathbb{C}/\Lambda$ and consider $n = 0$ as the trivial endomorphism sending the curve to the point 0 mod Λ. Note that the word endomorphism has to be understood here as indicating not only an endomorphism of the Riemann surface, but also an endomorphism of the group, fixing the point 0 mod Λ. Thus the covering transformation $z \bmod \Lambda \mapsto nz \bmod \Lambda$ inducing the quotient map $n^{-1}\mathcal{D} \to \mathcal{D}$ counts as an endomorphism, whereas the (Riemann surface automorphism) $z \bmod \Lambda \mapsto (z+a) \bmod \Lambda$ does not if $a \notin \Lambda$! This also helps to explain our slightly inconsistent notation, writing $\Gamma\backslash\mathbb{H}$ for quotients of the space $\mathbb{H}$, but $\mathbb{C}/\Lambda$ for quotients of the group $\mathbb{C}$, as is conventional in group theory.
If the endomorphism ring End T of T is strictly larger than $\mathbb{Z}$, we call T an **elliptic curve with complex multiplication**. This is the case if and only if the period quotient τ generates an imaginary quadratic extension K of $\mathbb{Q}$. If we denote the ring of integers of K by $\mathcal{O}_K$, and lift the endomorphisms to $\mathbb{C}$, then it is easy to see that End T is the subring of all $\alpha \in \mathcal{O}_K$ such that $\alpha\Lambda \subseteq \Lambda$. An automorphism of T occurs if α is a root of unity; since roots of unity generate imaginary quadratic fields if and only if they are of order 3, 4 or 6, we have the following:

Theorem 7. *A dessin $\mathcal{D}$ of genus 1 belongs to an elliptic curve with complex multiplication by $\mathbb{Z}\left[e^{2\pi i/3}\right]$ (or $\mathbb{Z}[i]$) if $\mathcal{D}$ has an automorphism of order 3 (or 4, respectively) fixing at least one vertex or face.*

Section 10.2.3 of [10] discusses dessins with this condition.

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